

Instructions:

- 1.The question paper contains two sections. Each section has three questions.
- 2.Each section carries 35 marks.
- 3.Figures to the right indicate marks.
- 4.Make suitable assumption whenever necessary and state them.

SECTION-I

Q-1 Attempt the following:

[10]

- (a) Define: Tautology.
- (b) Define: Principal disjunctive normal form.
- (c) Symbolize the expression "Every student of computer engineering has visited either Germany or Canada."
- (d) Let E_x : "x is an earth-like planet and L_x : "x supports life."
Give interpretation of $\sim (\forall x [E_x \vee L_x])$.
- (e) Let * be a Boolean operation defined by $A * B = AB + \bar{A}\bar{B}$, then $A * A$ is _____
- (f) Define Bounded Lattice.
- (g) Give an example of a nonempty, symmetric, transitive relation on the set $\{1, 2\}$ that is not reflexive.
- (h) What is the principle of duality?
- (i) Give an example of a minterm for a Boolean function of the three variables a, b and c .
- (j) Define Boolean algebra.

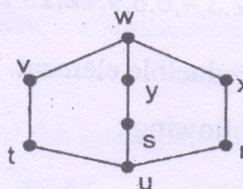
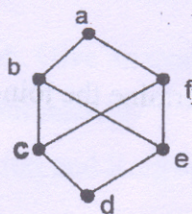
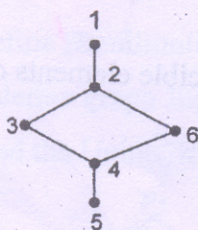
Q-2(a) Draw the Hasse diagram and find Maximal, Minimal, First and Last element of the

[04]

Poset $\{2,7,14,28,56,84\}$ with the relation $a \leq b$ if and only if 'a' divides 'b'.

- (b) Define Partially Ordered Set & Determine whether the posets represented by each of the following Hasse diagrams are Lattices or not by justifying your answer.

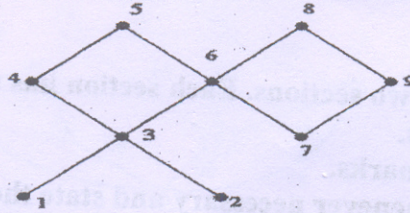
[04]



- (c) Show that $\sim q$ is a valid conclusion from the given premises $p \vee q \Rightarrow r$, $r \Rightarrow s$ & $\sim s$ **[03]**
- (d) Simplify the Boolean expression: $z \bullet (y + z) \bullet (x + y + z)$. **[02]**

OR

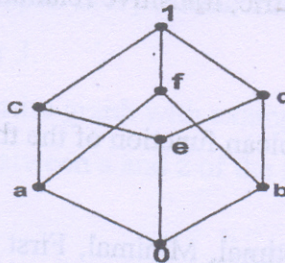
- Q-2(a) Consider the following Hasse diagram and the set $B = \{3, 4, 6\}$. Find, if they exist, (i) All upper bounds of B ; (ii) All lower bounds of B ; (iii) The least upper bound of B ; (iv) The greatest lower bound of B . [04]



- (b) Draw the Hasse diagram of the set of all divisors of 42 with respect to the division relation. Is this lattice bounded or not? Justify your answer. [04]
- (c) Find the principal conjunctive normal form of $p \vee (q \Rightarrow r)$. [03]
- (d) Write down converse and contra positive of the statement, If a triangle is isosceles, then the base angles are equal. [02]

- Q-3 (a) Define minterm and maxterm in Boolean algebra. Convert Product of Sum form $f(a, b, c) = (a + b + c) \cdot (b' + c') \cdot (a' + c')$ to canonical form. [05]

- (b) Consider the Boolean algebra $(B, +, \cdot, ', 0, 1)$ as shown in the figure. Let the subsets of B are: (i) $B_1 = \{c, b, 0, 1\}$ (ii) $B_2 = \{a, b, 0, 1\}$ (iii) $B_3 = \{a, b, 0, f\}$ (iv) $B_4 = \{a, c, f, 1\}$. Determine whether the B_1, B_2, B_3, B_4 are Boolean algebra, sub-Boolean algebra, either both or none. [04]

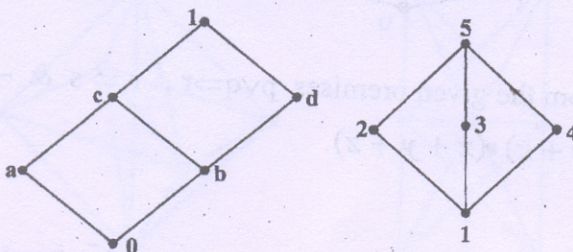


- (c) Minimize the Boolean expression : $a'c + a'b + ab'c + bc$. [03]

OR

- Q-3(a) What is the Karnaugh map? Using K-map, minimize the Boolean function $f(w, x, y, z) = \sum (0, 1, 2, 3, 4, 6, 8, 9, 12, 13, 14)$ [05]

- (b) Define atom and join-irreducible element. Determine the join-irreducible elements of the lattices as show in the following : [04]



- (c) Prove the following Boolean expression: $(x \vee y) \wedge (x \vee y') \wedge (x' \vee z) = x \wedge z$ [03]

SECTION-II

Q-4 Do as directed:

[07]

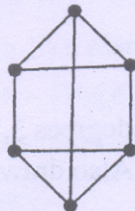
- (a) The number of vertices in a full binary tree is always _____.
- (b) In the adjacency matrix of a graph, the number of 1's in each column equals the degree of the corresponding vertex. True/False.
- (c) State Euler's theorem for planar graph.
- (d) Define Simple graph with example.
- (e) K_5 is a planar graph. True/False.
- (f) Define Fundamental circuit with example.
- (g) Determine the number of edges in a graph with 6 nodes 2 of degree 4 and 4 of degree 2. Draw such two graphs.

Q-5(a) Define Complement of graph. Draw the Complement of K_5 .

[02]

(b) Define Isomorphic graphs. Is the graph $K_{3,3}$ Isomorphic to the following graph

[03]



(c) Define Incident matrix, Adjacent matrix and draw graph of the following:

[05]

$$\begin{bmatrix} 1 & 1 & 2 \\ 1 & 0 & 1 \\ 2 & 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 & 1 & 0 \\ 1 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 1 & 1 & 1 \end{bmatrix}$$

(d) For a graph with 15 edges and each vertex has degree 3, determine the number of vertices in a graph. State the result used.

[02]

(e) Give example of following graph :

[02]

- (1) Both Eulerian circuit and Hamiltonian circuit.
- (2) An Eulerian circuit but **not** Hamiltonian circuit.

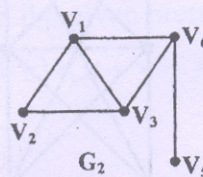
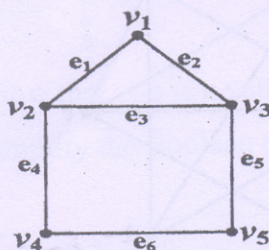
OR

Q-5(a) Define Hamiltonian graph. Give an example of a Hamiltonian graph which is not Eulerian graph Justify your answer.

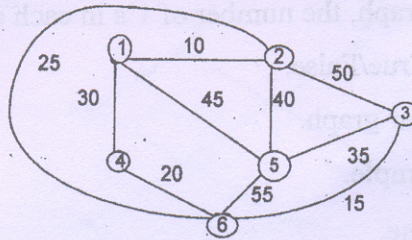
[03]

(b) Find the Union, Intersection and Ring sum of the given pair of graphs:

[03]



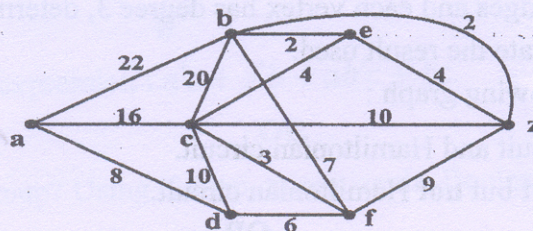
- (C) Find the minimal spanning tree by Kruskal algorithm for the following weighted graph [03]



- (d) Draw a graph having the given properties or explain why no such graph exists. [03]
- Simple graph with four vertices each of degree 2.
 - Seven vertices having degrees 3, 5, 2, 7, 4, 6 and 8.
- (e) Define Homomorphic graphs with example. [02]
- Q-6(a) Show that a Simple planar graph has a vertex of degree 5 or less. [03]
- Define Edge connectivity, Spanning tree, Rooted tree, Separable graph, Block with suitable example to each. [05]
 - Explain Stereographic projection. [03]
 - A Connected planar graph has seven vertices having degrees 3, 3, 2, 3, 3, 2 and 4. How many edges are there? How many regions are there? Also draw one such graph. [03]

OR

- Q-6(a) Define Tree with example. Is it possible to draw a tree with 11 vertices, which has vertices of degree 1 or 3. [03]
- Define Complete bipartite graph with example and draw $K_{1,1}$ and $K_{2,3}$. [03]
 - Find a Shortest path between a and z of the following weighted graph. [04]



- (d) State Kuratowski theorem. Determine which of the graphs shown in figure are planar or non planar. [04]

